

LUIS LOPEZ RONDON

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Education

University of South Florida

Bachelor of Science in Electrical Engineering

Tampa, FL

GPA: 4.00 | May 2029

Hillsborough College

Associate of Arts

Tampa, FL

GPA: 4.00 | May 2026

Relevant Coursework: Differential Equations, Digital Circuits

Work Experience

Tutoring Club of Westchase

Math and Reading Tutor

Tampa, FL

Jan 2026 – Aug 2026

- Tutored up to 3 K–12 students at a time across reading and mathematics curricula, ranging from elementary material through AP Calculus AB, SAT, and ACT.
- Helped students review mistakes, strengthen problem-solving strategies, and prepare for SAT practice material.

Leadership & Activities

IEEE at USF, EXO

Electronics and Circuitry Lead

Tampa, FL

2026 – Present

- Developing the electrical system for an exoskeleton leg driven by 48V CubeMars AK80-9 actuators, including IMU and FSR sensing and DC-DC rails for the Jetson Orin Nano and Teensy logic.
- Created a schematic for a hardware e-stop using a contactor relay to cut the motor bus while keeping control electronics powered, alongside a software e-stop for development testing.
- Performed a weighted benefit analysis of 48V battery options, comparing pre-built and custom packs on capacity, weight, size, discharge rating, and cost, and led design reviews to select a pack within budget.
- Designed a PCB using Altium Designer linking a flex-sensor glove to a servo-driven robotic hand in a mimetic hand project, taking it from breadboard prototype to board layout.
- Taught PCB design fundamentals to newer members and ran an Altium workshop where students designed a PCB with an ESP32-C3, IMU, and battery charging for a PID-stabilized Parkinson's spoon.

Mu Alpha Theta

Founder and President

Tampa, FL

Aug 2023 – May 2026

- Founded the school's Mu Alpha Theta chapter, launched tutoring initiatives, and built student interest in math competitions in Calculus, Precalculus, and Geometry.
- Tutored 20+ students to reach their PERT Benchmark, achieving a 90% pass rate.
- Individually tutored 10+ students in AP Calculus, AP Physics C, and SAT preparation.

Mu Alpha Theta Calculus Team

Captain

Tampa, FL

Aug 2023 – May 2026

- Organized team practices and developed practice problem sets to prepare members for competition.
- Led the team to place Top 5 in four different regional competitions, and individually placed top 20 at States.

Skills

Software: Altium Designer, KiCad, Keysight ADS, LTSpice

Microcontrollers: ESP32, Teensy, STM32

Hardware: PCB design, power distribution, DC-DC regulation, battery systems, e-stop and safety circuitry

Languages: English, Spanish

Awards & Recognition

National Merit Finalist, Sunshine State Scholar, 1530 SAT, Mu Alpha Theta Regional Competitions Top 5